

WASP-43b

R-0378

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250406 · created 2026-07-18T18:53:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460771.7119 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.047
Transit depth (Rp/Rs)^2	2.48 +/- 1.49 [%]
Orbital Inclination (inc)	86.09 +/- 2.69
Airmass coefficient 1 (a1)	0.846 +/- 0.014
Airmass coefficient 2 (a2)	0.12 +/- 0.12
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	#1 - [82, 118]
Optimal Aperture	3.72
Optimal Annulus	10.09
Transit Duration (day)	0.0578 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.047 vs 0.1594 (deviation 0.05σ)
Duration fit vs published	0.0578 d vs 0.0512 d (deviation 1.24σ)
Mid-time O–C	4.1 min
Depth SNR	3.3
Residual scatter	1.6 %

Light curve

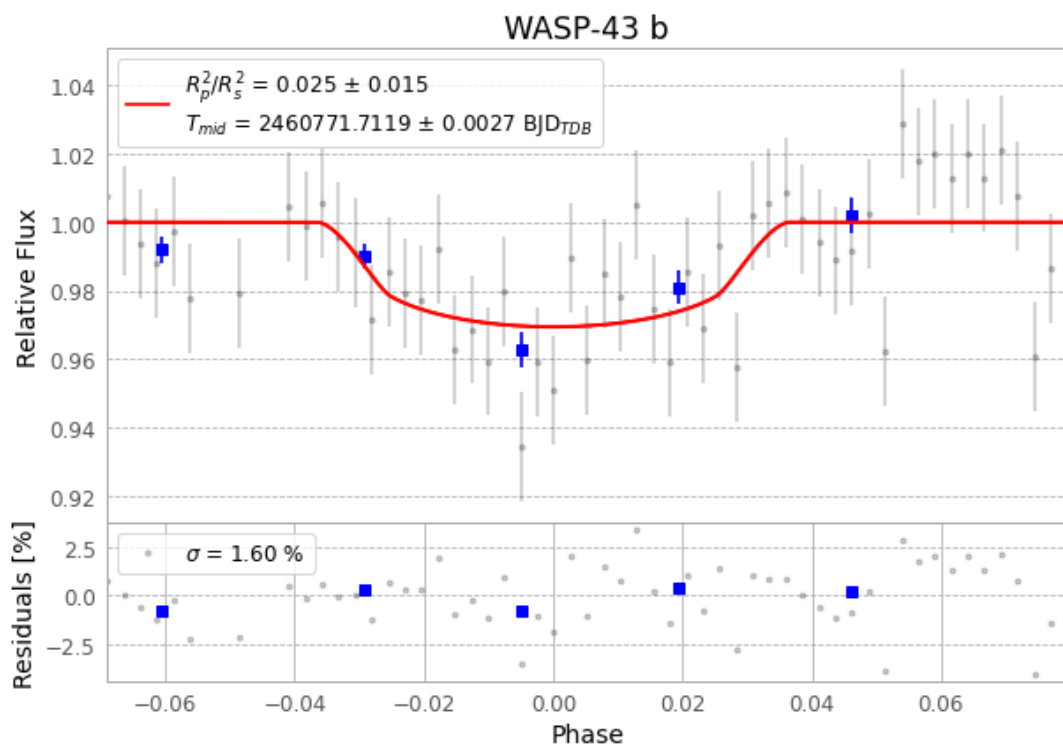


Figure 1 — FinalLightCurve_WASP-43 b_2025-04-05.png (SHA-256 54b5e84059e3933a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f41a53c81c7d940a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

59 FITS frames (39 MB), WASP-43250406033615.FITS → WASP-43250406063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250406.log (cfbd9e02d296...), FinalLightCurve_WASP-43 b_2025-04-05.png (54b5e84059e3...), FinalLightCurve_WASP-43 b_2025-04-05.csv (6838966b2077...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 18:53Z.